ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** magnesium	Basic oxide, so metal.
2	**(d)** magnesium	Magnesium is highly reactive.
3	**(b)** zinc	Acid reacts.
4	**(d)** sulphur	Non-metal.
5	**(a)** Fe	Iron.
6	**(c)** copper	Good conductor.
7	**(a)** zinc	Zinc.
8	**(a)** blue to green	$CuSO_4$ is blue.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Malleability is the property of being beaten into thin sheets, so R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but being a form of carbon does not explain why graphite conducts.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Zinc sulphate and hydrogen (1).	1+1
12	They dissolve in rain to form acids (1); Such as SO_2 giving sulphurous acid (1).	1+1
13	They conduct heat away from the hand (1); Heat flows quickly (1).	1+1
14	Gold is very unreactive (1); Silver tarnishes with sulphur (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Metal (1); Magnesium (1); Basic oxide (1).	1+1+1
16	$MgO + H_2O \longrightarrow Mg(OH)_2$ (1); $SO_2 + H_2O \longrightarrow H_2SO_3$ (1); Basic and acidic (1).	1+1+1
17	They are brittle or soft (1); Hammers need hard metal (1); Iron or steel used (1).	1+1+1
	OR: Colour (1); Density (copper is heavier) (1); Conductivity (1).	1+1+1
18	Soft (1); Reactive with air (1); Surface tarnishes (1).	1+1+1
	OR: Sodium soft, copper hard (1); Sodium very reactive (1); Copper conducts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Metals are good conductors of heat (1). (ii) Plastic is a poor conductor of heat, so the handle does not become hot (1). (iii) Malleability (1). (iv) Ductility (1); copper is also a good conductor of electricity (1).	1+1+1+2
20	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); $Mg + CuSO_4 \longrightarrow MgSO_4 + Cu$ (1); Mg is reactive (1); Products observed (1).	1+1+1+1+1
	OR: Copper (1); Bronze alloy (1); Iron (1); Steel (1); Modern alloys (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** increases	Copper heavier.
2	**(b)** less reactive	They do not displace hydrogen.
3	**(a)** carbon	CO2.
4	**(d)** copper	Brown.
5	**(d)** copper sulphate	Copper ions replaced.
6	**(a)** both are forms of carbon	Allotropes.
7	**(b)** $ZnSO_4$	Iron is less reactive than zinc.
8	**(a)** making electric wires	Wires are of metals.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Iron takes the place of copper in the solution and copper is deposited on the nail.
10	**(d)** A is false but R is true.	Mercury is a liquid metal, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tungsten conducts and has high melting point (1); Sulphur is an insulator and melts easily (1).	1+1
12	Copper is less reactive than hydrogen (1); No displacement (1).	1+1
13	Iron is strong and stable (1); Sodium is too reactive and soft (1).	1+1
14	Good conductors (1); Carry charge to ground (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It may be an unreactive metal (1); Protective oxide coating (1); Acid may be too dilute (1).	1+1+1
16	Oxide layer present (1); Prevents burning (1); Sand paper removes it (1).	1+1+1
17	Iron displaces copper (1); Iron sulphate is green (1); Copper deposits (1).	1+1+1
	OR: Magnesium + oxygen gives magnesium oxide (1); Magnesium oxide is a white powder (1); It is basic in nature (1).	1+1+1
18	Iron rusts in water and air (1); Coating keeps water and air away (1); Tank lasts longer (1).	1+1+1
	OR: Zinc, iron, magnesium (1); Sodium reacts violently (1); Safety (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Magnesium oxide (1). (ii) It turns red litmus blue, as it is basic (1). (iii) Sulphur dioxide (1). (iv) It turns blue litmus red (1); this shows that oxides of non-metals are acidic (1).	1+1+1+2
20	Oxygen for respiration (1); Carbon in organic compounds (1); Nitrogen in fertilisers (1); Sulphur in medicines (1); Chlorine disinfects (1).	1+1+1+1+1
	OR: Predict properties (1); Select materials (1); Study reactions (1); Organise knowledge (1); Exceptions noted (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** it does not support burning	It is used in fire extinguishers.
2	**(c)** metal	Sonorous.
3	**(c)** calcium oxide	$CaO + H_2O \longrightarrow Ca(OH)_2$.
4	**(c)** salt and water	Neutralisation.
5	**(a)** graphite	An exceptional non-metal.
6	**(c)** sulphur dioxide	Non-metal oxide.
7	**(a)** lead	Poor among metals.
8	**(b)** different reactions with air	Different products.
9	**(c)** A is true but R is false.	Magnesium oxide is basic (turns red litmus blue) so A is true; metal oxides are basic, so R is false.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but heat conduction has nothing to do with the reaction with acids.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Forms copper oxide (1); Black coating (1).	1+1
12	Add the oxide solution to turmeric (1); It turns red (1).	1+1
13	To protect them from rusting (1); The coating keeps air and moisture away (1).	1+1
14	Liquid metal (1); Expands evenly (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Steel is stronger (1); Iron rusts easily (1); Steel lasts longer (1).	1+1+1
16	Metals expand on heating (1); Wire becomes longer (1); Longer wire hangs lower (1).	1+1+1
17	It does not support burning (1); Heavier than air (1); Covers the fire (1).	1+1+1
	OR: Move magnet over the mixture (1); Iron filings stick (1); Sulphur is left (1).	1+1+1
18	Zinc protects iron from rusting (1); The zinc layer keeps out air and moisture (1); Such iron is called galvanised (1).	1+1+1
	OR: Mixtures of metals (1); Brass: copper and zinc (1); Steel: iron and carbon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Reddish brown (copper) (1). (ii) Blue to pale green (1). (iii) A displacement reaction (1). (iv) No (1), because copper is less reactive than iron and cannot displace it (1).	1+1+1+2
20	Tests show metals light the bulb (1); Wax melts along a metal rod (1); Non-metals do not (1); Graphite is an exception (1); Use decides (1).	1+1+1+1+1
	OR: Saves ores (1); Saves energy (1); Reduces waste (1); Reduces pollution (1); Creates jobs (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

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